Applied Machine Learning - Basic

Prof. Daniele Bonacorsi, Dott. Luca Clissa

Lecture 0

[05.03.2026]

Data Science and Computation PhD + Master in Bioinformatics
University of Bologna

Lingua / Language ?

Italiano / English ?

Lingua / Language ?

Italiano / **English**

Lectures and material in English

Introduction and Logistics

Material (including these slides)

Access point to all you need → **Virtuale** for the course

https://bit.ly/AMLBas2526

People

This part (**Applied ML - Basic**):

Prof. **Daniele Bonacorsi** (daniele.bonacorsi@unibo.it)

Dr. **Luca Clissa** (luca.clissa2@unibo.it)

- Both: front lectures + hands-on

Next part (**Applied ML - Advanced**):

Prof. **Daniele Bonacorsi** (daniele.bonacorsi@unibo.it)

Dr. **Luca Clissa** (luca.clissa2@unibo.it)

- Both: front lectures + hands-on

Who am I?

https://www.unibo.it/sitoweb/daniele.bonacorsi

I am a **physicist**. My field is "**experimental high-energy particle physics (HEP)**", in particular with particle accelerators.

Research:

- subnuclear physics in OPAL at LEP, in CMS at LHC
- over last >20 years: focus on Software/Computing for the CMS experiment
  - CERN: https://home.cern/; CMS: https://cms.cern/

Teaching:

- General Physics, Data Analysis and Physics Lab, Applied Machine Learning, Software&Computing (focus on Nuclear and Subnuclear Physics), Quantum Machine Learning, general introductions to Artificial Intelligence
- "Scuole" (i.e. Faculties): Physics, Engineering, Natural Sciences, Bioinformatics
  - L, LM, Master (on HPC and Quantum), PhD
- (currently) Coordinator of the PhD in "Data Science and Computation"

What this course **is**, and what it **is not**

First of all, "expectations tuning".

Despite we will talk about…

- ML concepts, statistics, data science, computer science, software, hardware..

… this course:

- is <u>not</u> a "theoretical" ML / statistics course
- is <u>not</u> a "theoretical" {data, computer} science course

It is **a ~30 + ~50 hours journey through (selected) ML concepts and their application to a variety of use-cases through "intuitions" from theory + tutorial-like hands-on exercises**

- allow me brevity, accelerations, simplifications, time-constraints driven choices

Consider it a "portal" towards ***<u>your</u>*** travel into data science and ML

- you should expect to build some awareness + skills on toolkits, that enable you to embark into some "real world" ML projects during/after the end of the course

"Sito docente" → "Didattica"

ITA: https://www.unibo.it/sitoweb/daniele.bonacorsi/didattica
ENG: https://www.unibo.it/sitoweb/daniele.bonacorsi/teachings

B5803 - APPLIED MACHINE LEARNING-BASIC - 4 cfu

Campus: Bologna

Degree programme: Second cycle degree programme (LM) in Bioinformatics
Second cycle degree programme (LM) in Bioinformatics

Lesson period: from March 5, 2026 to April 10, 2026

Online Lessons | Course timetable | Teaching resources on Virtuale

B5779 - APPLIED MACHINE LEARNING (Module 1) - 10 cfu

Campus: Bologna

Degree programme: Second cycle degree programme (LM) in Bioinformatics

Lesson period: from March 5, 2026 to April 10, 2026

Course timetable | Teaching resources on Virtuale

B5779 - APPLIED MACHINE LEARNING - 10 cfu

Campus: Bologna

Degree programme: Second cycle degree programme (LM) in Bioinformatics

Lesson period: from March 5, 2026 to May 14, 2026

Course timetable | Teaching resources on Virtuale

B5799 - APPLIED MACHINE LEARNING-ADVANCED - 6 cfu

Campus: Bologna

Degree programme: Second cycle degree programme (LM) in Bioinformatics
Second cycle degree programme (LM) in Bioinformatics

Lesson period: from April 16, 2026 to May 14, 2026

Course timetable | Teaching resources on Virtuale

Historically, services quite unable to fix this in a proper and clean way. Be aware that this is NOT the best way to keep in touch with the course.. we need to organise ourselves independently.

Web site for the course

https://www.unibo.it/en/study/course-units-transferable-skills-moocs/course-unit-catalogue/course-unit/2025/524681

NOTE: I am displaying here the FULL course, all modules, 10 CFU - AML Basic is part of it (4 CFU)

B5779 - APPLIED MACHINE LEARNING

Academic Year 2025/2026

Docente: Daniele Bonacorsi **Credits:** 10 **SSD:** FIS/01 **Language:** English

Moduli: Daniele Bonacorsi (Modulo 1) Daniele Bonacorsi (Modulo 2) Luca Clissa (Modulo 3)

Teaching Mode: In-person learning (entirely or partially) (Modulo 1); In-person learning (entirely or partially) (Modulo 2); In-person learning (entirely or partially) (Modulo 3)

Campus: Bologna

Corso: Second cycle degree programme (LM) in Bioinformatics (cod. 6767)

Also valid for Second cycle degree programme (LM) in Bioinformatics (cod. 6767)
Second cycle degree programme (LM) in Bioinformatics (cod. 8020)

Course Timetable
from Mar 05, 2026 to Apr 10, 2026

Course Timetable
from Apr 16, 2026 to May 14, 2026

Course Timetable
from May 08, 2026 to May 29, 2026

Calendar [1/2]

Lesson period Mar 05, 2026 - Apr 10, 2026

Thursday, 5 March 2026	14:00 - 18:00	1F Piano Primo U.E. 1 - via della beverara, 123/1 Via della beverara, 123/1 - Bologna
Friday, 6 March 2026	14:00 - 18:00	1F Piano Primo U.E. 1 - via della beverara, 123/1 Via della beverara, 123/1 - Bologna
Thursday, 12 March 2026	14:00 - 18:00	
Friday, 13 March 2026	14:00 - 18:00	
Friday, 27 March 2026	14:00 - 18:00	1F Piano Primo U.E. 1 - via della beverara, 123/1 Via della beverara, 123/1 - Bologna
Tuesday, 31 March 2026	14:00 - 18:00	1F Piano Primo U.E. 1 - via della beverara, 123/1 Via della beverara, 123/1 - Bologna
Wednesday, 1 April 2026	09:00 - 13:00	1F Piano Primo U.E. 1 - via della beverara, 123/1 Via della beverara, 123/1 - Bologna
Friday, 10 April 2026	14:00 - 18:00	1F Piano Primo U.E. 1 - via della beverara, 123/1 Via della beverara, 123/1 - Bologna

NOTE: I am displaying here only the AML Basic part

IMPORTANT: Dates are susceptible to change. In case, you will be notified in advance via **Virtuale**.

Calendar [2/2]

Admittedly: tough. Details apart, you note that:

- always 4 hours in a row
- Alway afternoon, until late

I will do my best to mitigate everything for your protection.

Lectures format

Format:

- Sort of "**hybrid**":
  - **lectures are in presence** (mandatory for Bioinformatics students),
    - I will make some use of Teams → more info live at the introductory lecture
- **Frontal** vs **hands-on**
  - **frontal lectures** to start..
  - .. then (later on) we will inject **hands-on sessions**
- every hourly slot will be in the format: **45 mins lecture +15 mins pause**
  - this is my choice to support your best fruition of the course - if you prefer it different, speak up!
- **ask questions!**

FAQs on lectures and material

Material available? Yes.

- you will have slides (hopefully in advance, on Virtuale)
- PDF updates *after* each lecture are possible, so check back!

Mandatory **attendance**? I am not asking it

Remote attendance?

- For all students in Bioinformatics, lectures should be in presence
- For PhDs and special cases → comment at the lecture

Recordings? No.

- I may activate it for special needs, of for internal usage (e.g. tutors' training), not for students

Course material and organisation

Static and dynamic material

- **Slides** and **Jupyter/Colab** notebooks
- (brief comment at the lecture about copyrights and material usage)

Material both on **Virtuale** and **GitHub**

- Gdrive: — look for the course —
- GitHub: — as soon as we start using notebooks —

Access to course material:

- during the course: open to everyone having the link
- after the course: might restrict access to material from old courses

Course details (as from the web site)

https://www.unibo.it/en/study/course-units-transferable-skills-moocs/course-unit-catalogue/course-unit/2025/524681

In the next few slides, we will go through **selected** parts of the web page above.

It is your duty to READ CAREFULLY the full page,
and ask questions in case of doubts!

Learning outcomes (as from the web site)

B5803 - APPLIED MACHINE LEARNING-BASIC

Academic Year 2025/2026

Learning outcomes

At the end of the course the student is able to manage the main concepts of Machine Learning and apply them in examples, from data preparation to model selection and training. This course is propedeutic to "Applied Machine Learning - Advanced", where the student is introduced to techniques and methods used in a variety of more advanced applications of Machine/Deep Learning in real use cases.

Modules (as from the web site)

Course contents

The course consist of traditional (frontal) lectures and practical sessions (i.e. hands-on activities on topics covered in the course) on these subjects.

Lectures on:

- Definitions of Machine Learning
- Types of Machine Learning
- Univariate and Multivariate Linear Regression
- Cost functions
- Gradient Descent and Learning Rate
- Alternatives to Gradient Descent
- Feature scaling, standardisation, normalisation
- Polynomial regression
- Computing parameters analytically: the normal equation
- Classification and representation
- Hypothesis representation and Decision boundaries
- Logistic regression and related cost function
- Multiclass classification
- The problem overfitting
- Regularisation and modified cost functions
- Regularised linear regression, regularised logistic regression
- ML model selection and evaluation
- Static train/validation/test splitting vs cross-validation
- Bias vs Variance diagnosis; learning curves
- Error metrics for skewed classes
- Confusion matrix, Precision, Recall, F1, ROC, AUC

Hands-on activities on:

- Jupyter / Google Colab(oratory)
- Data preparation: loading a dataset
- Data preparation: exploring a dataset
- Data preparation: preparing a dataset for ML
- Data preparation: feature selection
- Data preparation: evaluation with resampling
- Modelling: algorithms' performance metrics
- Modelling: model comparison
- Modelling: algorithm tuning and performance improvements

Teaching methods (as from the web site)

Teaching methods

The teaching methodology of the course includes:

- Traditional frontal lectures;
- Interactive notebooks for the practical sessions that involve coding.

All of this is complemented by innovative teaching approaches, which may include the use of Microsoft Teams for remote access, online materials (e.g., videos), and the organization of "flipped classroom" experiences on specific topics.

The slides and notebooks used during the lessons, along with any other materials shown in class, are fully made available to students. In particular, the slides for the lectures are provided to students before the corresponding class, allowing them to take notes directly on the course material.

For the hands-on sessions, the availability of computer rooms is not guaranteed, so students are advised to bring their own laptops.

Exam rules (as from the web site)

Here: in general. Later: details for Bioinformatics LM vs PhD students vs Erasmus

Assessment methods

The exam consists of two parts, each contributing equally to the final grade (15 points each, for a total of 30).

1) WRITTEN TEST (15 points out of 30)

- **Objective**: Assess the students' understanding and conscious, accurate elaboration of the concepts taught during the lectures.
- **Participation in the assessment:** To take the written test, students must register via the *Almaesami* website (almaesami.unibo.it) by the indicated deadlines, typically a few days before the scheduled date.
- **Delivery method:** The written test is held in person. Exceptions that allow remote participation via Microsoft Teams are considered only if a justified and documented request is submitted to the instructors, and only if official certification verifying the impediment is received by the course coordinator no later than one week before the scheduled exam date.
- **Description and evaluation of the assessment:** The test consists of a *multiple-choice questionnaire*, administered either through EOL (eol.unibo.it) or on paper. The questions are of equivalent difficulty, and each one carries the same weight. Correct answers award positive points; incorrect answers may result in negative points, at the instructors' discretion (who may decide to apply or not apply penalties for wrong answers, for a specific session or the entire academic year). The test *duration* may range from a minimum of 1 hour to a maximum of 2 hours, depending on the number and difficulty of the questions. To ensure fairness and inclusiveness, the test is never time-pressured: students are always given ample time to reflect and answer all questions calmly — on average, at least 5 minutes per question. During the test, the use of a calculator and any printed reference material (notes, textbooks, handouts, etc.) is *allowed*, including teaching materials shown and distributed during lectures. However, the use of any device or method allowing communication outside the exam room (whether with people or digital tools), as well as any form of communication between students during the exam, is *strictly prohibited*. Violation of these rules will result in immediate annulment of the test. The written test can be taken multiple times by registering for future sessions on Almaesami, with the goal of improving the score. The highest score obtained is always retained as the valid one. Each score remains valid for one calendar year; if it expires, the student must retake the written test.

Exam rules (as from the web site)

Here: in general. Later: details for Bioinformatics LM vs PhD students vs Erasmus

2) PROJECT WORK (15 points out of 30)

- **Objective:** Assess the student's ability to apply the course content to a coding-based project.
- **Participation in the assessment:** No registration via Almaesami is required.
- **Delivery method:** The project may be chosen by the student, either by submitting their own proposal or by selecting one from a list provided by the instructors. Students must inform the instructors of the project they intend to work on, including a brief description, and must receive instructor approval before proceeding.
- **Description and evaluation of the assessment:** Project submission guidelines and timelines are presented during the first lecture of the course. Evaluation is based primarily on: clarity in explaining the goals and the approach used to achieve them; correctnes in the ML methodology and quality of the implemented code; documentation and overall presentation of the project. Each project may be submitted up to two times. After the first submission, the student will receive feedback and a proposed score. The student can either accept the score or decide to improve the project based on the feedback and submit it once more. If the score of the second submission is not accepted, the same project cannot be submitted a third time: the student must propose a new project from scratch. In the case of special needs (e.g., deadlines for scholarships or graduation sessions), the submission modalities and deadlines may be discussed on an individual basis with the instructors, who will make every effort to accommodate the student within the bounds of fair and equal treatment for all.

Exam rules (as from the web site)

The maximum score a student can obtain is 15+15=30. Honors (“30L”) may be awarded to students who obtain the maximum score in both components of the exam and whose overall performance is unanimously considered excellent by the instructors.

Written exams session (summer 2026): tentative dates

We are planning to have:

- *(Perhaps!) a "partial" in-course written test (on AML Basic content only)*
- 3 dates in June/July
- 1 date in September
- 2 dates in January/February

Tentatively, for the summer 2026 (all TO BE CONFIRMED yet!)

- Around June 18th or 19th
- Around July 2nd or 3rd
- Around July 23rd or 24th

When confirmed, **you will see them on Almaesami**

- Note you can sign up only later on, but dates will be visible soon

More on the exam

Exam: who does what?

Bioinformatics LM students:

- Written test
  - (sign-up on Almaesami, execution on EOL, need to bring a laptop)
- Project work

PhD young researchers (e.g. Data Science and Computation, Physics, etc.)

- Project work

Erasmus students:

- → discussion (and agreement) at the lectures

Guests

- Doors are open. No exams. Just attend and enjoy

For any other case : please contact the teachers' team.

Exam: common questions

How long will my obtained score last?

Can I take another written test?

My written test expired, is it still valid?

What is actually a "project"?

Can I have a third project submission?

Can I show you the project before submitting it?

Can I do Bas only, or Adv only?

Can I do a single project for both Bas and Adv?

Can I do a single written test for Bas and Adv altogether?

I do not have the course in my study plan: can I give the exam?

Can I do the project work with a friend at the course?

How long should I wait for a correction of my project?

How long should I wait before seeing the final, accepted grade inserted on Almaesami?

Answers in class at the lecture. But most answers should already be in the previously mentioned web site for the course (please report what's missing, so we can improve).

More on communication

How to communicate

Info from us to you

- Virtuale

Questions from you to us

- Always mail **both** professors
  - Prof. **Daniele Bonacorsi** (daniele.bonacorsi@unibo.it)
  - Dr. **Luca Clissa** (luca.clissa2@unibo.it)

My contacts

daniele.bonacorsi@unibo.it

NOTE: to avoid unreceived/unseen mails, please ALWAYS write to all teachers + tutor

@DBonacorsi

NOTE: I used Twitter pretty much, I am not using X as much.